Evaluation of Mechanical Properties of Recycled AA6061-Copper $_{(P)}$ Composite Metallic Materials

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Abstract -The recycling of aluminum scraps and translation of useful product has concerned more and more thoughtfulness of researchers in order to reduce emission and waste disposal. The composite material is coming under the high-performance materials system and has several advantages over the conventional materials owing to its superior properties such as greater tensile, impact and fatigue properties. The aforementioned properties are considered for aerospace, automotive and ship construction industries. The present paper deals the manufacturing of aluminum composite metallic materials by using scrap aluminum and copper particulates via stir casting. The composite samples prepared with reinforcing ratio of the 2%, 4% and 6%. The mechanical properties such as hardness, tensile and impact properties are evaluated. It is found from the analysis, the increment in copper reinforcement rises the entire mechanical properties.

INTRODUCTION

A huge quantity of scrap aluminum scraps is produced in foundries. The effort on recycling the aluminum alloy maximizes its recovery value and minimize the material shortage thereby reduce the cost of aluminum alloy [1]. Manufacturing of the composite material by using the scrap aluminum alloy one of the novel ways to recycle the aluminum alloys. The recycling effort on scraped aluminum alloy reduce existing mechanical property. However, the manufacturing of the composite material one of effective way to regain the mechanical property of the scrap aluminum alloy. The composites materials fabricated by combining the two considerably distinct materials in order to suit the specific applications. The properties of the composites can be tailored by varying the quantity of reinforcing phase. Bulei et al [2] addressed the challenges associated with the manufacturing composite material by using the recycled aluminum wastes and noted that the recycling of aluminum is valued in terms of economically and ecologically. Christy et al [3] observed that there is no new phase formation in the composite materials when using the recycled Aluminum as the matrix material. The analysis of the experimental investigation discloses that the achievement of superior mechanical properties at the optimum processing condition. The following literatures deals the development of composites by reinforcing copper particulates. Akinribide et al [4] found that the tensile strength of the composite samples was certainly influenced by the adding of copper phase in terms of evenly distributed reinforcing intermetallic phases with fine grain size in addition to a noticeable reduction of micro-segregation. Thus, enhancing the mechanical properties of the alloys of the developed composites. Madhusudan et al [5] evaluated the mechanical properties of the Al-Cu $_{(p)}$ composite metallic materials under various volume contents of Cu. Their study reveals that the mechanical properties of the composites increased up to 10 % Cu than decreased due to excessive agglomeration of Cu. The literature review illustrates that the recycling of the aluminum alloy mostly encouraged for avoiding material shortage. The usage as a resource of waste generated from foundries and the reduction of the amount of waste that reaches landfills are primary goal. Addition of Cu alloy and manufacturing the Al-Cu $_{(p)}$ composite metallic materials is an effective strategy for recycling the scrap aluminum. Therefore, the current work fills the gap of manufacturing the Al-Cu $_{(p)}$ composite metallic materials and evaluating the mechanical properties. The objective of the work is to manufacture the Al-Cu $_{(p)}$ composite metallic materials reinforcing ratio of the 2%, 4% and 6%. Further, the mechanical properties such as hardness, tensile and impact properties are to be evaluated.

MATERIALS AND METHOD

The scrap aluminum material collected from the foundries. The aluminum alloy used in the current experiment is Al 6061. Mg and Si are major alloying elements present in the alloy and it have significant mechanical properties including corrosion resistance, toughness and deformation behavior. The alloy exhibits greater weldability and machinability and it is widely used for general-purpose. Copper has greater conductivity and corrosion resistance together with higher hardness. It is good reinforcing phase to improve the mechanical properties of the recycled aluminum and it helps to retain the raw aluminum properties. There are 3 samples are prepared reinforcing ratio of the 2%, 4% and 6% by using the stir casting. The various steps associated with the sample manufacturing is described below.

Coal fired furnace is used for casting. A crucible is preheated up to 300 º C and placed the scrap aluminum 6061 alloy into the crucible. The heat applied up to 700 º C and allowed the alloy reaches in liquid form. The slag formed due to the presence of oil, oxide layer and other dust on the scrap aluminum alloy is removed. The copper powders are kept in a pan and preheated up to 300 º C in order to remove the moisture and other contaminations. The preheated copper particulates are added and stirred up 20 minutes. The continuous stirring helps to maintain the uniform distribution of the copper particulates within the aluminum matrix. The molten mixture is then poured in preheated mold and allowed for air cooling and solidification. The photographic view of the composite samples is illustrated in the Figure. 1.

FIGURE 1: PHOTOGRAPHIC VIEW OF COMPOSITE SAMPLES

EXPERIMENTAL WORK

From the fabricated composites, the specimens for tensile, hardness and impact tests were made as per ASTM stand. There are 3 samples are prepared reinforcing ratio of the 2%, 4% and 6% A Computerized universal testing machine is used for the measuring the yield strength and ultimate tensile test. ASTME8 standard is followed for the tensile test. The specimens for impact test also prepared from the samples as per ASTM E23 stand. Impact Strength tests were performed by Charpy impact testing machine. The dimensions of the specimens were 10 mm x 10mm and 55 mm. The hardness of samples is evaluated by Brinell hardness test. Scale used: B-Scale with Hardened Steel ball indenter used for the hardness test. The diameter of the indenter is 1.5mm. The major load is

100 kg and minor load is 10 kg.

RESULT AND DISCUSSION

The yield and tensile strength of the composite samples are evaluated and illustrated in the Table.1. The photographic view of the tensile samples before and after the test are illustrated in the Figure 2 and 3 receptively. The Table 1 depicts that the yield strength of the AA6061- 2 % Cu sample has lower than other samples. Furthermore, the AA6061- 6 % Cu has greater yield strength among the samples. It is also noted from the Table 1, the yield strength is increased by increasing the coper percentage. The ultimate tensile strength also followed the same trend as seen from yield strength. AA6061- 6 % Cu has greater tensile strength and the ultimate tensile strength is increased by raising the coper percentage.

TABLE 1: Tensile Strength of The Composite Samples

Sample	**Yield strength (MPa)**	**Ultimate Tensile Strength (MPa)**
AA6061- 2 % Cu	77.33	112
AA6061- 4 % Cu	88	114.67
AA6061- 6 % Cu	98.6	120

FIGURE 2: TENSILE SPECIMENS A) BEFORE TEST B) AFTER TEST

The impact strength of the composites are evaluated and illustrated in the Table 2. The photographic view of the samples after impact test also illustrated in the Figure 3. It is clear from the Table 2, the AA6061- 6 %Cu has greater impact strength and the impact strength is increased by raising the coper percentage. The impact strength of the 6 % Cu reinforced composite has 6.87 Jules whereas 2 % Cu reinforced composite has 4.95 joules.

TABLE 2: Impact Strength

Sample	**Impact strength (Joules)**
AA6061- 2 % Cu	4.95
AA6061- 4 % Cu	5.85
AA6061- 6 % Cu	6.87

FIGURE 3: IMPACT TEST SPECIMENS AFTER TEST

The hardness tests are carried in the composite samples and illustrated in the Table 3. The average hardness number of the sample with 2 % Cu has 55 whereas the 6 % has 60. Further, the Table 3 also depicts that the increment in copper percentage linearly increasing the hardness.

TABLE 3: Hardness Test

Sample	Trail-1 (BHN)	Trail-2 (BHN)	Mean (BHN)
AA6061- 2 % Cu	57	53	55
AA6061- 4 % Cu	54	61	57.5
AA6061- 6 % Cu	64	56	60

The analysis of the results shows that the clear raise in all mechanical properties while increasing the copper percentage. The Al_2Cu phase formation plays vital role in increasing the mechanical behavior of the materials. However, the AA 6061-O has ultimate tensile strength of 150 MPa and yield strength has 86 MPa [6]. The recycled aluminum after addition of Cu reinforcement, the yield strength is almost regained when compared to the yield strength of AA 6061-O. Conversely, the ultimate tensile strength of the 2 % Cu, 4 % Cu and 6 % Cu reinforced samples were reached 74 %, 77 % and 81 % of the AA 6061-O has ultimate tensile strength. It is clear from the analysis; the recycling effect of aluminum alloy decline the ultimate tensile strength of AA6061alloy even after reinforcing the Cu alloy in recycled AA6061 matrix. Therefore, further investigations are required to enhance the mechanical properties during recycling of aluminum alloy.

CONCLUSION

- The present work deals the manufacture the Al-Cu (p) composite metallic materials reinforcing ratio of the 2%, 4% and 6%. Further, the mechanical properties such as hardness, tensile and impact properties are to be evaluated and discussed. From the experimental based investigation, the following conclusions are attained.
- The yield strength and ultimate tensile strength of the composite samples are increased by raising the percentage of copper. The Al_2Cu phase formation plays vital role in increasing the mechanical behavior of the materials.
- The greater impact strength is achieved at 6 % Cu reinforced composite samples. It is interesting to note that the impact strength of the composite samples is increased by raising the percentage of copper. The reduction in impact strength may be reduced greater than 10 % Cu.
- Hardness of the composite samples are raised by increasing copper percentage.
- The ultimate tensile strength of the 2 % Cu, 4 % Cu and 6 % Cu reinforced samples were reached 74%, 77 % and 81 % of the ultimate tensile strength AA 6061-O alloys.

- The recycling effect of aluminum alloy decline the ultimate tensile strength of AA6061 alloy even after reinforcing the Cu alloy in recycled AA6061 matrix

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